Clinton Enwerem

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Summary

Robotics Research Engineer with 5+ years of research experience and Electrical and Computer Engineering Ph.D. candidate at the University of Maryland, graduating in Fall 2026. Develop demonstration pipelines, evaluate imitation/RL policies, and implement Python/C++ sensing and control software for dexterous manipulation in simulation and on physical robots.

Skills

Robotics: cuRobo, ROS/ROS 2, ros2_control, MoveIt 2, MoveIt Servo, SocketCAN, VR teleoperation; **Planning & Control:** RRT*, FCL, state estimation, CBF-CLF QPs, MPC; **Learning:** PyTorch, OpenVLA-7B, QLoRA, ACT, DAgger, Diffusion Policy, PPO, flow matching; **Simulation:** Isaac Lab/Sim, Gazebo, MuJoCo, Drake; **Software:** Python, C++, Linux, Docker, git, CI; **GPU Computing:** CUDA, Warp, MuJoCo Warp, JAX; **Perception:** FoundationPose, BundleSDF, SAM 2, SAM 3D, PCL, OpenCV; **Robots:** Fairino FR3, xArm7, UR3e, UR5, Robotiq Hand-E, RealHand A7 (Bimanual Arm-Hand System), RealHand L6, LEAP Hand v1, Unitree G1 (simulation).

Work Experience

Institute for Systems Research (ISR), University of Maryland Aug. 2021 – Present
Graduate Research Assistant *College Park, MD*
• Develop grasp-synthesis, state-estimation, motion-planning, and feedback-control methods for dexterous manipulation under contact uncertainty; implement Python/C++ and ROS 2 software for physical arm–hand systems.
• Trained PPO with GDF rewards in **1,024 Isaac Lab environments**: **64/64** pregrasp arrivals versus **0/64** with palm-distance rewards.
• Developed xArm7-RealHand-L6 VR teleoperation with wrist-to-Cartesian velocity mapping, relative joint-position control, grasp primitives, thumb abduction, and hand reset; integrated MoveIt Servo with Dimensional OS (DimOS) for collision detection, arm speed modulation, and velocity damping near kinematic boundaries.
• Fine-tuned OpenVLA-7B with QLoRA for recovery supervision in simulated bimanual packing; each of three training seeds achieved 136/200 task completions versus 129/200 without supervision, recovering seven failures without disrupting nominal successes.
• Developed a visuo-tactile bimanual assembly pipeline with simulation and hardware execution interfaces, combining depth-based pose estimation (RANSAC/DBSCAN, ICP), collision-constrained trajectory optimization (CasADi/IPOPT), and gravity-compensated position control with fingertip-contact monitoring. Evaluated task execution in MuJoCo.

MATRIX Lab, USM at Southern Maryland Jun. – Aug. 2022
Research Intern *California, MD*
• Identified a discrete-time Crazyflie model from flight data and implemented a finite-horizon optimal formation controller in Python and Crazyswarm; executed leader–follower formation flights with 10 physical Crazyflie quadrotors under localization uncertainty.

Kognitive Robotics Mar. 2021 – Aug. 2021
Robotics Engineer *Lagos, Nigeria*
• Designed power and sensor–actuator interfaces in Altium; assembled a mobile robot and developed and tested ROS software for the platform.

Robotics & Artificial Intelligence Nigeria (RAIN) Jan. 2020 – Feb. 2021
Robotics Trainee *Ibadan, Nigeria*
• Developed a ROS-based quadrotor autonomy stack on a Raspberry Pi 3, integrating ORB-SLAM3 visual SLAM, an EKF fusing SLAM poses with IMU data, and a PID flight controller. The onboard controller maintained stable hover during ground-station packet loss.

Projects

ParcelStow — Execution-Timing Robustness in Imitation Learning (Paper, project, Code, Data & Models) | *Isaac Lab, PyTorch* 2026
• Built ParcelStow, an Isaac Lab benchmark for assessing imitation-policy robustness to execution-timing variation; generated **970K** expert states (**297** demonstrations) and trained ACT, Diffusion Policy, and DAgger. Controlled timing stress tests exposed precision-insertion failures: at 2× execution speed, ACT achieved **53%** success versus **84%** for the expert.

Grasp Distance Fields — Feedback Grasp Execution (Project, Paper) | *C++, Python, QP* 2026
• Implemented a CBF-CLF quadratic-program controller for feedback grasp execution; simulated experiments lifted **46 of 50** objects, with a median QP solve time of **0.09 ms**.

dexreach — GPU-Accelerated Dexterous Grasp and Motion Planning (Project) | *cuRobo, PyTorch, Isaac Lab* 2026
• Built GPU-accelerated grasp synthesis and cuRobo motion planning with friction/actuator checks and Isaac Lab validation: dual Franka-Ability completed both objects' sequences in **6/6** tested plans, with **4–19 mm** final position errors.

Selected Publications

• **C. Enwerem et al.** "Does Imitation Learning Preserve Temporal Robustness in Dexterous Manipulation? An Expert–Learner Comparison Across Task Execution Speeds." *arXiv preprint*, 2026. arXiv:2609.01453, project page, code, data & models.
• **C. Enwerem et al.** "Grasp Execution Without a Planner: Configuration-Space Grasp Distance Fields with Certified Safety & Guaranteed Quality." *Preprint*, 2026. arXiv:2608.00600, project page.
• **C. Enwerem et al.** "Safety-Aware Reinforcement Learning for Control via Risk-Sensitive Action-Value Iteration and Quantile Regression." *CDC 2025*. arXiv:2506.06954.
• Additional first-author papers in control, safe learning, and robotic manipulation: IROS 2026; CDC 2026/2025/2024. Full list: Google Scholar.

Education

Ph.D., Electrical & Computer Engineering, University of Maryland, College Park, MD. Expected Fall 2026. Aug. 2021 – Present
B.Eng., Electrical Engineering, University of Nigeria, Nsukka. Highest Honors. GPA 3.84/4.0. Aug. 2018